

Ensemble

Machine Learning Implementation for Network Intrusion Detection

> Essential ML process for Intrusion Detection

python 3.7.13 scikit-learn 1.0.2

numpy 1.19.5 pandas 1.3.5

Import the main libraries

```
In [11]: class="keyword">import numpy
class="keyword">import pandas

class="keyword">from time class="keyword">import time

class="keyword">import os
data_path = class="string">'../data' class="string">"comment">#
Updated for Assignment folder structure
```

import the local library

```
In [12]: class="string">"comment"># add parent folder path where lib folder
is
class="keyword">import sys
if class="string">".." not in sys.path:class="keyword">import sys;
sys.path.insert(class="number">0, class="string">'..')
```

```
In [13]: class="keyword">from mylib class="keyword">import show_labels_dist,
show_metrics, bias_var_metrics
```

Import the Dataset

```
In [14]: class=class="string">"comment"># Using boosted Train and preprocessed Test

data_file = os.path.join(data_path, class="string">'NSL_boosted-
class="number">2.csv')
train_df = pandas.read_csv(data_file)
print(class="string">'Train Dataset: {} rows, {}
columns'.format(train_df.shape[class="number">0],
train_df.shape[class="number">1]))

data_file = os.path.join(data_path, class="string">'NSL_ppTest.csv')
test_df = pandas.read_csv(data_file)
print(class="string">'Test Dataset: {} rows, {}
columns'.format(test_df.shape[class="number">0],
test_df.shape[class="number">1]))
```

```
Train Dataset: 63280 rows, 43 columns
Test Dataset: 22544 rows, 43 columns
```

*

Data Preparation and EDA (unique to this dataset)

- _Check column names of numeric attributes_

```
In [15]: trnn = train_df.select_dtypes(include=
[class="string">'float64',class="string">'int64']).columns
tstn = test_df.select_dtypes(include=
[class="string">'float64',class="string">'int64']).columns
trndif = numpy.setdiff1d(trnn, tstn)
tstdif = numpy.setdiff1d(tstn, trnn)

print(class="string">"Numeric features in the train_set that are not in
the test_set: ",end=class="string">')
if len(trndif) > class="number">0:
    print(class="string">'\n',trndif)
else:
    print(class="string">'None')

print(class="string">"Numeric features in the test_set that are not in the
train_set: ",end=class="string">')
if len(tstdif) > class="number">0:
    print(class="string">'\n',tstdif)
else:
    print(class="string">'None')

print()
class=class="string">"comment"># correct any differences here
```

```
Numeric features in the train_set that are not in the test_set: None
Numeric features in the test_set that are not in the train_set: None
```

- _Check column names of categorical attributes_

```
In [16]: trnn = train_df.select_dtypes(include=[class="string">'object']).columns
tstn = test_df.select_dtypes(include=[class="string">'object']).columns
trndif = numpy.setdiff1d(trnn, tstn)
tstdif = numpy.setdiff1d(tstn, trnn)

print(class="string">"Categorical features in the train_set that are not
in the test_set: ",end=class="string">'')
if len(trndif) > class="number">0:
    print(class="string">'\\n',trndif)
else:
    print(class="string">'None')

print(class="string">"Categorical features in the test_set that are not in
the train_set: ",end=class="string">'')
if len(tstdif) > class="number">0:
    print(class="string">'\\n\\t',tstdif)
else:
    print(class="string">'None')

print()
class=class="string">"comment"># correct any differences here
```

```
Categorical features in the train_set that are not in the test_set: None
Categorical features in the test_set that are not in the train_set: None
```

- _Check for missing values_

```

In [17]: cnt=class="number">0
print(class="string">'Missing Values - Train Set')
for col in train_df.columns:
class=class="string">"comment">#      print(col, class="string">' :> ',
len(combined_df[col].unique()))
    nnul = pandas.notnull(train_df[col])
    if (len(nnul)!=len(train_df)):
        cnt=cnt+class="number">1
        print(class="string">'\\t',col,class="string">':',(len(test_df)-
len(nnul)),class="string">'null values')
print(class="string">'Total',cnt,class="string">'features with null
values')

cnt=class="number">0
print(class="string">'Missing Values - Test Set')
for col in test_df.columns:
class=class="string">"comment">#      print(col, class="string">' :> ',
len(combined_df[col].unique()))
    nnul = pandas.notnull(test_df[col])
    if (len(nnul)!=len(test_df)):
        cnt=cnt+class="number">1
        print(class="string">'\\t',col,class="string">':',(len(test_df)-
len(nnul)),class="string">'null values')
print(class="string">'Total',cnt,class="string">'features with null
values')

class=class="string">"comment"># address missing values here

```

```

Missing Values - Train Set
Total 0 features with null values
Missing Values - Test Set
Total 0 features with null values

```

- _Quick visual check of unique values, deal with unique identifiers_

In [18]:

```
class=class="string">"comment"># Identify columns with only one value
class=class="string">"comment"># or with number of unique values == number
of rows
n_eq_one = []
n_eq_all = []

print(class="string">'Unique value count:
Train(',train_df.shape[class="number">0],class="string">'rows ) ~
Test(',test_df.shape[class="number">0],class="string">'rows )')
for col in train_df.columns:
    lctrn = len(train_df[col].unique())
    lctst = len(test_df[col].unique())
    print(col, class="string">' ::> ', lctrn, class="string">' ~ ', lctst)
    if (lctrn == class="number">1) and (lctrn == lctst):
        n_eq_one.append(train_df[col].name)
    if lctrn == train_df.shape[class="number">0]:
        n_eq_all.append(train_df[col].name)
```

```
Unique value count: Train ( 63280 rows ) ~ Test( 22544 rows )
duration ::> 1657 ~ 624
protocol_type ::> 3 ~ 3
service ::> 70 ~ 64
flag ::> 11 ~ 11
src_bytes ::> 2553 ~ 1149
dst_bytes ::> 6633 ~ 3650
land ::> 2 ~ 2
wrong_fragment ::> 3 ~ 3
urgent ::> 3 ~ 4
hot ::> 24 ~ 16
num_failed_logins ::> 6 ~ 5
logged_in ::> 2 ~ 2
num_compromised ::> 49 ~ 23
root_shell ::> 2 ~ 2
su_attempted ::> 3 ~ 3
num_root ::> 44 ~ 20
num_file_creations ::> 28 ~ 9
num_shells ::> 3 ~ 4
num_access_files ::> 9 ~ 5
num_outbound_cmds ::> 1 ~ 1
is_host_login ::> 2 ~ 2
is_guest_login ::> 2 ~ 2
count ::> 511 ~ 495
srv_count ::> 489 ~ 457
serror_rate ::> 87 ~ 88
srv_serror_rate ::> 75 ~ 82
rerror_rate ::> 81 ~ 90
srv_rerror_rate ::> 63 ~ 93
same_srv_rate ::> 101 ~ 75
diff_srv_rate ::> 88 ~ 99
srv_diff_host_rate ::> 61 ~ 84
dst_host_count ::> 256 ~ 256
dst_host_srv_count ::> 256 ~ 256
dst_host_same_srv_rate ::> 101 ~ 101
dst_host_diff_srv_rate ::> 101 ~ 101
dst_host_same_src_port_rate ::> 101 ~ 101
dst_host_srv_diff_host_rate ::> 67 ~ 58
dst_host_serror_rate ::> 101 ~ 99
```

```
dst_host_srv_error_rate ::> 100 ~ 101
dst_host_rerror_rate ::> 101 ~ 101
dst_host_srv_rerror_rate ::> 101 ~ 100
label ::> 40 ~ 38
atakatcat ::> 5 ~ 5
```

```
In [19]: class=class="string">"comment"># Drop columns with only one value
if len(n_eq_one) > class="number">0:
    print(class="string">'Dropping single-valued features')
    print(n_eq_one)
    train_df.drop(n_eq_one, axis=class="number">1, inplace=True)
    test_df.drop(n_eq_one, axis=class="number">1, inplace=True)

class=class="string">"comment"># Drop or bin columns with number of unique
values == number of rows
if len(n_eq_all) > class="number">0:
    print(class="string">'Dropping unique identifiers')
    print(n_eq_all)
    train_df.drop(n_eq_all, axis=class="number">1, inplace=True)
    test_df.drop(n_eq_all, axis=class="number">1, inplace=True)

class=class="string">"comment"># continue with feature selection / feature
engineering
```

```
Dropping single-valued features
['num_outbound_cmds']
```

- _Check categorical feature values:

differences will be resolved by one-hot encoding the combined test and train sets_

```

In [20]: trnn = train_df.select_dtypes(include=[class="string">'object']).columns
for col in trnn:
    tr = train_df[col].unique()
    ts = test_df[col].unique()
    trd = numpy.setdiff1d(tr, ts)
    tsd = numpy.setdiff1d(ts, tr)

    print(col, class="string">'::> ')
    print(class="string">"\tUnique text values in the train_set that are
not in the test_set: ", end=class="string">'')
    if len(trd) > class="number">0:
        print(class="string">'\\n\\t', trd)
    else:
        print(class="string">'None')

    print(class="string">"\tUnique text values in the test_set that are
not in the train_set: ", end=class="string">'')
    if len(tsd) > class="number">0:
        print(class="string">'\\n\\t', tsd)
    else:
        print(class="string">'None')

```

```

protocol_type ::>
    Unique text values in the train_set that are not in the test_set:
None
    Unique text values in the test_set that are not in the train_set:
None
service ::>
    Unique text values in the train_set that are not in the test_set:
['aol' 'harvest' 'http_2784' 'http_8001' 'red_i' 'urh_i']
    Unique text values in the test_set that are not in the train_set:
None
flag ::>
    Unique text values in the train_set that are not in the test_set:
None
    Unique text values in the test_set that are not in the train_set:
None
label ::>
    Unique text values in the train_set that are not in the test_set:
['spy' 'warezclient']
    Unique text values in the test_set that are not in the train_set:
None
atakcat ::>
    Unique text values in the train_set that are not in the test_set:
None
    Unique text values in the test_set that are not in the train_set:
None

```

- _Combine for processing classification target and text features_

```
In [21]: combined_df = pandas.concat([train_df, test_df])
print(class="string">'Combined Dataset: {} rows, {} columns'.format(
    combined_df.shape[class="number">0],
    combined_df.shape[class="number">1]))
```

Combined Dataset: 85824 rows, 42 columns

- _Classification Target feature:_

two columns of labels are available

* Two-class: Reduce the detailed attack labels to 'normal' or 'attack'

* Multiclass: Use the category labels (atakat)


```
In [22]: combined_df[class="string">'label'].value_counts()
```

```
Out[22]: label
normal          43383
neptune         25264
satan           2552
smurf           1988
ipsweep         1941
portsweep       1623
guess_passwd    1257
mscan           1046
warezmaster     954
back            837
nmap            820
apache2         774
processtable    719
teardrop        458
warezclient     445
snmpguess       364
saint           350
mailbomb        322
snmpgetattack   196
httptunnel      146
pod             142
buffer_overflow 35
named           25
ps              22
multihop        21
sendmail        21
xterm           20
rootkit         18
land            16
xlock           14
xsnoop          8
ftp_write       7
imap            6
loadmodule      6
phf             6
perl            5
worm            4
sqlattack       4
udpstorm        4
spy             1
Name: count, dtype: int64
```

```
In [23]: combined_df[class="string">'atakat'].value_counts()
```

```
Out[23]: atakat
benign    43383
dos       30524
probe     8332
r2l       3329
u2r       256
Name: count, dtype: int64
```

```
In [24]: class=class="string">"comment"># Set the classification target
twoclass = True      class=class="string">"comment"># True or False
```

```
In [25]: if twoclass:
class=class="string">"comment"># Two-class: Reduce the detailed attack
labels to class="string">'normal' or class="string">'attack'
class=class="string">"comment"># new single column data structure is a
[series]
    labels_df = combined_df[class="string">'label'].copy()
    labels_df[labels_df != class="string">'normal'] =
class="string">'attack'
else:
class=class="string">"comment"># Multiclass: Use the category
labels(atakat)
class=class="string">"comment"># new single column data structure is a
[[dataframe]]
class=class="string">"comment"># rename the column and convert to a series
for later
    labels_df = combined_df[[class="string">'atakat']].copy()
    labels_df.rename(columns=
{class="string">'atakat':class="string">'label'}, inplace=True)
    labels_df = labels_df.squeeze(class="string">'columns')

class=class="string">"comment"># drop target features
combined_df.drop([class="string">'label'], axis=class="number">1,
inplace=True)
combined_df.drop([class="string">'atakat'], axis=class="number">1,
inplace=True)
```

- _One-Hot Encoding the categorical (text) features_

```
In [26]: class=class="string">"comment"># put the names into a python list - for
pandas.get_dummies()
categori = combined_df.select_dtypes(include=
[class="string">'object']).columns
category_cols = categori.tolist()
print(category_cols)
```

```
['protocol_type', 'service', 'flag']
```

```
In [27]: class=class="string">"comment"># generate a sorted list of unique values
of categorical features
class=class="string">"comment"># we will get a new column for each one
with get_dummies()

for col in categori:
    ul = numpy.sort(list(combined_df[col].unique()))
    print(col, class="string">' ::> ', ul)
    print()
```

```
protocol_type ::> ['icmp' 'tcp' 'udp']

service ::> ['IRC' 'X11' 'Z39_50' 'aol' 'auth' 'bgp' 'courier' 'csnet_ns'
'ctf'
'daytime' 'discard' 'domain' 'domain_u' 'echo' 'eco_i' 'ecr_i' 'efs'
'exec' 'finger' 'ftp' 'ftp_data' 'gopher' 'harvest' 'hostnames' 'http'
'http_2784' 'http_443' 'http_8001' 'imap4' 'iso_tsap' 'klogin' 'kshell'
'ldap' 'link' 'login' 'mtp' 'name' 'netbios_dgm' 'netbios_ns'
'netbios_ssn' 'netstat' 'nnsdp' 'nntp' 'ntp_u' 'other' 'pm_dump' 'pop_2'
'pop_3' 'printer' 'private' 'red_i' 'remote_job' 'rje' 'shell' 'smtp'
'sql_net' 'ssh' 'sunrpc' 'supdup' 'systat' 'telnet' 'tftp_u' 'tim_i'
'time' 'urh_i' 'urp_i' 'uucp' 'uucp_path' 'vmnet' 'whois']

flag ::> ['OTH' 'REJ' 'RSTO' 'RSTOS0' 'RSTR' 'S0' 'S1' 'S2' 'S3' 'SF'
'SH']
```

```
In [28]: class=class="string">"comment"># Apply to the list of Categorical
columns(text fields)
features_df = pandas.get_dummies(combined_df, columns=category_cols)
features_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 85824 entries, 0 to 22543
Columns: 121 entries, duration to flag_SH
dtypes: bool(84), float64(15), int64(22)
memory usage: 31.8 MB
```

```
In [29]: class=class="string">"comment"># generate a list of numeric columns for
scaling - After test // train split
numeri = combined_df.select_dtypes(include=
[class="string">'float64',class="string">'int64']).columns
print(numeri.to_list())
```

```
['duration', 'src_bytes', 'dst_bytes', 'land', 'wrong_fragment', 'urgent',
'hot', 'num_failed_logins', 'logged_in', 'num_compromised', 'root_shell',
'su_attempted', 'num_root', 'num_file_creations', 'num_shells',
'num_access_files', 'is_host_login', 'is_guest_login', 'count',
'srv_count', 'error_rate', 'srv_error_rate', 'rerror_rate',
'srv_rerror_rate', 'same_srv_rate', 'diff_srv_rate', 'srv_diff_host_rate',
'dst_host_count', 'dst_host_srv_count', 'dst_host_same_srv_rate',
'dst_host_diff_srv_rate', 'dst_host_same_src_port_rate',
'dst_host_srv_diff_host_rate', 'dst_host_error_rate',
'dst_host_srv_error_rate', 'dst_host_rerror_rate',
'dst_host_srv_rerror_rate']
```

*

Create Test // Train Datasets

> Normally we split the dataset into train 70 % // test 30 % like this

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test =
```

```
train_test_split(features_df, labels_df,
```

```
test_size=0.3, stratify=labels_df, random_state=42)
```

```

In [30]: class=class="string">"comment"># Restore the train // test split: slice
class="number">1 Dataframe into class="number">2
features_train = features_df.iloc[:len(train_df),:].copy()
class=class="string">"comment"># X_train
features_train.reset_index(inplace=True, drop=True)
class=class="string">"comment"># pandas has a lot of rules about returning
a class="string">'view' vs. a copy class="keyword">from slice
class=class="string">"comment"># so we force it to create a new dataframe
[avoiding SettingWithCopy Warning]
features_test = features_df.iloc[len(train_df)::].copy()
class=class="string">"comment"># X_test
features_test.reset_index(inplace=True, drop=True)

class=class="string">"comment"># Restore the train // test split: slice
class="number">1 Series into class="number">2
labels_train = labels_df[:len(train_df)]
class=class="string">"comment"># y_train
labels_train.reset_index(inplace=True, drop=True)

labels_test = labels_df[len(train_df):]
class=class="string">"comment"># y_test
labels_test.reset_index(inplace=True, drop=True)

```

*

Next are standard steps for all datasets: `_scaling`, `classifiers`, `results_`

Scaling comes `_after_` `test // train split`

```

In [31]: class=class="string">"comment"># scaling the Numeric columns
class=class="string">"comment"># StandardScaler range: -class="number">1
to class="number">1, MinMaxScaler range: zero to class="number">1

class=class="string">"comment"># class="keyword">from
sklearn.preprocessing class="keyword">import StandardScaler
class="keyword">from sklearn.preprocessing class="keyword">import
MinMaxScaler

class=class="string">"comment"># sklearn docs say
class=class="string">"comment"># class="string">"Don't cheat - fit only
on training data, then transform both"
class=class="string">"comment"># fit() expects 2D array: reshape(-
class="number">1, class="number">1) for single col or (class="number">1, -
class="number">1) single row

for i in numeri:
    arr = numpy.array(features_train[i])
    scale = MinMaxScaler().fit(arr.reshape(-class="number">1,
class="number">1))
    features_train[i] =
scale.transform(arr.reshape(len(arr),class="number">1))

    arr = numpy.array(features_test[i])
    features_test[i] =
scale.transform(arr.reshape(len(arr),class="number">1))

```

Classifier Selection

In [32]:

```
class=class="string">"comment"># prepare list
models = []

class=class="string">"comment">## -- Linear -- ##
class=class="string">"comment">#class="keyword">from sklearn.linear_model
class="keyword">import LogisticRegression
class=class="string">"comment">#models.append((class="string">"LogReg",Log
isticRegression()))
class=class="string">"comment">#class="keyword">from sklearn.linear_model
class="keyword">import SGDClassifier
class=class="string">"comment">#models.append((class="string">"StocGradDes
",SGDClassifier()))
class=class="string">"comment">#class="keyword">from
sklearn.discriminant_analysis class="keyword">import
LinearDiscriminantAnalysis
class=class="string">"comment">#models.append((class="string">"LinearDA",
LinearDiscriminantAnalysis()))

class=class="string">"comment">## -- Support Vector -- ##
class=class="string">"comment">#class="keyword">from sklearn.svm
class="keyword">import SVC
class=class="string">"comment">#models.append((class="string">"SupportVect
orClf", SVC()))

class=class="string">"comment">## -- Non-linear -- ##
class=class="string">"comment">#class="keyword">from sklearn.tree
class="keyword">import DecisionTreeClassifier
class=class="string">"comment">#models.append((class="string">"DecisionTre
e",DecisionTreeClassifier()))
class=class="string">"comment">#class="keyword">from sklearn.neighbors
class="keyword">import KNeighborsClassifier
class=class="string">"comment">#models.append((class="string">"K-
NNeighbors", KNeighborsClassifier()))

class=class="string">"comment">## -- Ensemble: bagging -- ##
class="keyword">from sklearn.ensemble class="keyword">import
RandomForestClassifier
models.append((class="string">"RandomForest", RandomForestClassifier()))
class="keyword">from sklearn.ensemble class="keyword">import
BaggingClassifier
models.append((class="string">"BaggingClf", BaggingClassifier()))
class="keyword">from sklearn.ensemble class="keyword">import
ExtraTreesClassifier
models.append((class="string">"ExtraTrees", ExtraTreesClassifier()))

class=class="string">"comment">## -- Ensemble: boosting -- ##
class=class="string">"comment">#class="keyword">from sklearn.ensemble
class="keyword">import AdaBoostClassifier
class=class="string">"comment">#models.append((class="string">"AdaBoost",
AdaBoostClassifier()))
class=class="string">"comment">#class="keyword">from sklearn.ensemble
class="keyword">import GradientBoostingClassifier
class=class="string">"comment">#models.append((class="string">"GradientBoo
st", GradientBoostingClassifier()))

print(models)
```

```
[('RandomForest', RandomForestClassifier()), ('BaggingClf',
BaggingClassifier()), ('ExtraTrees', ExtraTreesClassifier())]
```

`_compatibility` block for pasting in from sample code_

```
In [33]: class=class="string">"comment"># dataset names
X_train = features_train
y_train = labels_train
X_test = features_test
y_test = labels_test
labels_col = class="string">'label'
class=class="string">"comment"># library names
pd = pandas
np = numpy
```

Target Label Distributions (standard block)

```
In [34]: class=class="string">"comment"># class=keyword>from our local library
show_labels_dist(X_train,X_test,y_train,y_test)
```

```
features_train: 63280 rows, 121 columns
features_test:  22544 rows, 121 columns
```

```
labels_train: 63280 rows, 1 column
labels_test:  22544 rows, 1 column
```

```
Frequency and Distribution of labels
      label  %_train  label  %_test
label
normal  33672    53.21   9711   43.08
attack  29608    46.79  12833   56.92
```

`_... this is a good place for the ClassBalance visualizer ..._`

Fit and Predict (standard block)


```

In [35]: class=class="string">"comment"># evaluate each model in turn
results = []

print(class="string">'macro average: unweighted mean per label')
print(class="string">'weighted average: support-weighted mean per label')
print(class="string">'MCC: correlation between prediction and ground
truth')
print(class="string">'      (class="number">1 perfect, class="number">0
random prediction, -class="number">1 inverse)\n')

for name, clf in models:
    trs = time()
    print(class="string">'Confusion Matrix:', name)

    clf.fit(X_train, y_train)
    ygx = clf.predict(X_test)
    results.append((name, ygx))

    tre = time() - trs
    print (class="string">"Run Time {}
seconds".format(round(tre,class="number">2)) + class="string">'\\n')

class=class="string">"comment"># Easy way to ensure that the confusion
matrix rows and columns
class=class="string">"comment">#   are labeled exactly class="keyword">as
the classifier has coded the classes
class=class="string">"comment">#   [[note the _ at the end of clf.classes_
]]

    show_metrics(y_test, ygx, clf.classes_)
class=class="string">"comment"># class="keyword">from our local library
    print(class="string">'\\nParameters: ', clf.get_params(),
class="string">'\\n\\n')

```

```

macro average: unweighted mean per label
weighted average: support-weighted mean per label
MCC: correlation between prediction and ground truth
      (+1 perfect, 0 random prediction, -1 inverse)

```

```

Confusion Matrix: RandomForest
Run Time 6.58 seconds

```

	pred:attack	pred:normal
train:attack	10527	2306
train:normal	431	9280

~~~~

```

macro avg : FPR = 0.112 FNR = 0.112
weighted avg : FPR = 0.121 FNR = 0.121

```

~~~~

	precision	recall	f1-score	support
attack	0.961	0.820	0.885	12833
normal	0.801	0.956	0.871	9711
accuracy			0.879	22544

macro avg	0.881	0.888	0.878	22544
weighted avg	0.892	0.879	0.879	22544

~~~~

MCC: Overall : 0.769

Parameters: {'bootstrap': True, 'ccp\_alpha': 0.0, 'class\_weight': None, 'criterion': 'gini', 'max\_depth': None, 'max\_features': 'sqrt', 'max\_leaf\_nodes': None, 'max\_samples': None, 'min\_impurity\_decrease': 0.0, 'min\_samples\_leaf': 1, 'min\_samples\_split': 2, 'min\_weight\_fraction\_leaf': 0.0, 'monotonic\_cst': None, 'n\_estimators': 100, 'n\_jobs': None, 'oob\_score': False, 'random\_state': None, 'verbose': 0, 'warm\_start': False}

Confusion Matrix: BaggingClf

Run Time 6.19 seconds

|              |             |             |
|--------------|-------------|-------------|
|              | pred:attack | pred:normal |
| train:attack | 10938       | 1895        |
| train:normal | 398         | 9313        |

~~~~

macro avg :	FPR = 0.094	FNR = 0.094
weighted avg :	FPR = 0.102	FNR = 0.102

~~~~

|              |           |        |          |         |
|--------------|-----------|--------|----------|---------|
|              | precision | recall | f1-score | support |
| attack       | 0.965     | 0.852  | 0.905    | 12833   |
| normal       | 0.831     | 0.959  | 0.890    | 9711    |
| accuracy     |           |        | 0.898    | 22544   |
| macro avg    | 0.898     | 0.906  | 0.898    | 22544   |
| weighted avg | 0.907     | 0.898  | 0.899    | 22544   |

~~~~

MCC: Overall : 0.804

Parameters: {'bootstrap': True, 'bootstrap_features': False, 'estimator': None, 'max_features': 1.0, 'max_samples': 1.0, 'n_estimators': 10, 'n_jobs': None, 'oob_score': False, 'random_state': None, 'verbose': 0, 'warm_start': False}

Confusion Matrix: ExtraTrees

Run Time 7.34 seconds

	pred:attack	pred:normal
train:attack	10492	2341
train:normal	808	8903

~~~~

|                |             |             |
|----------------|-------------|-------------|
| macro avg :    | FPR = 0.133 | FNR = 0.133 |
| weighted avg : | FPR = 0.140 | FNR = 0.140 |

~~~~

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

attack	0.928	0.818	0.870	12833
normal	0.792	0.917	0.850	9711
accuracy			0.860	22544
macro avg	0.860	0.867	0.860	22544
weighted avg	0.870	0.860	0.861	22544

~~~~

MCC: Overall : 0.727

Parameters: {'bootstrap': False, 'ccp\_alpha': 0.0, 'class\_weight': None, 'criterion': 'gini', 'max\_depth': None, 'max\_features': 'sqrt', 'max\_leaf\_nodes': None, 'max\_samples': None, 'min\_impurity\_decrease': 0.0, 'min\_samples\_leaf': 1, 'min\_samples\_split': 2, 'min\_weight\_fraction\_leaf': 0.0, 'monotonic\_cst': None, 'n\_estimators': 100, 'n\_jobs': None, 'oob\_score': False, 'random\_state': None, 'verbose': 0, 'warm\_start': False}

## Bias - Variance Decomposition (standard block)

```
In [37]: class=class="string">"comment"># class="keyword">from our local library
class=class="string">"comment"># reduce(cross-validation) folds for faster
results
folds = class="number">20
for name, clf in models:
    print(class="string">'Bias // Variance Decomposition:', name)
    bias_var_metrics(X_train,X_test,y_train,y_test,clf,folds)
```

```
Bias // Variance Decomposition: RandomForest
Average bias: 0.119
Average variance: 0.009
Average expected loss: 0.122 "Goodness": 0.878
```

```
Bias // Variance Decomposition: BaggingClf
Average bias: 0.110
Average variance: 0.031
Average expected loss: 0.108 "Goodness": 0.892
```

```
Bias // Variance Decomposition: ExtraTrees
Average bias: 0.140
Average variance: 0.012
Average expected loss: 0.138 "Goodness": 0.862
```

\*

\*

## **Baseline Model**

>Select this block - Go to the Run menu - Run all Above

Then paste in blocks below from the other examples  
and run them one at a time

\*

\*

## **Hyperparameter Tuning**

General pattern:

\_baseline model\_

> 1. Classifier selection

2. Fit and Predict

3. Bias-Variance Tradeoff

\_optimised model\_

> 4. Select strategy and hyperparameters

5. Plug in the best parameter values

6. Fit and Predict

7. Bias-Variance Tradeoff

\*

In [39]:

```
class=class="string">"comment">#
=====
class=class="string">"comment"># HYPERPARAMETER TUNING - ENSEMBLE BAGGING
CLASSIFIERS
class=class="string">"comment">#
=====
class=class="string">"comment"># KEY IMPROVEMENTS:
class=class="string">"comment">#   class="number">1.
class_weight=class="string">'balanced' via base estimator(handles class
imbalance)
class=class="string">"comment">#   class="number">2. Multiple tuning
configurations tested
class=class="string">"comment">#   class="number">3. Bias-variance
analysis for tuned model
class=class="string">"comment">#
=====

class="keyword">import matplotlib.pyplot class="keyword">as plt
class="keyword">import seaborn class="keyword">as sns
class="keyword">from sklearn.metrics class="keyword">import
balanced_accuracy_score, f1_score, accuracy_score, matthews_corrcoef
class="keyword">from sklearn.metrics class="keyword">import
confusion_matrix
class="keyword">from sklearn.ensemble class="keyword">import
BaggingClassifier, RandomForestClassifier, ExtraTreesClassifier
class="keyword">from sklearn.tree class="keyword">import
DecisionTreeClassifier

print(class="string">"="*class="number">70)
print(class="string">"HYPERPARAMETER TUNING - ENSEMBLE CLASSIFIERS")
print(class="string">"="*class="number">70)
print(class="string">"\nClass Distribution(Note: Test set differs
class="keyword">from Train!):")
print(fclass="string">"  Train:
{(y_train=='normal').mean()*class="number">100:.1f}% normal,
{(y_train=='attack').mean()*class="number">100:.1f}% attack")
print(fclass="string">"  Test:
{(y_test=='normal').mean()*class="number">100:.1f}% normal,
{(y_test=='attack').mean()*class="number">100:.1f}% attack")
```

```
=====
HYPERPARAMETER TUNING - ENSEMBLE CLASSIFIERS
=====

Class Distribution (Note: Test set differs from Train!):
  Train: 53.2% normal, 46.8% attack
  Test:  43.1% normal, 56.9% attack
```

In [40]:

```
class=class="string">"comment">#
=====

class=class="string">"comment"># STEP class="number">1: BASELINE
COMPARISON - All Ensemble Classifiers
class=class="string">"comment">#
=====

print(class="string">"\n" + class="string">="*class="number">70)
print(class="string">"STEP class="number">1: BASELINE COMPARISON")
print(class="string">="*class="number">70)

baseline_models = [
    (class="string">"RandomForest",
RandomForestClassifier(n_estimators=class="number">100,
random_state=class="number">42, n_jobs=-class="number">1)),
    (class="string">"BaggingClassifier",
BaggingClassifier(n_estimators=class="number">100,
random_state=class="number">42, n_jobs=-class="number">1)),
    (class="string">"ExtraTrees",
ExtraTreesClassifier(n_estimators=class="number">100,
random_state=class="number">42, n_jobs=-class="number">1))
]

all_results = {}
for name, clf in baseline_models:
    print(fclass="string">"\nTraining {name}...")
    trs = time()
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)
    tre = time() - trs

    all_results[name] = {
        class="string">'accuracy': accuracy_score(y_test, y_pred),
        class="string">'balanced_accuracy':
balanced_accuracy_score(y_test, y_pred),
        class="string">'f1_score': f1_score(y_test, y_pred,
average=class="string">'weighted'),
        class="string">'mcc': matthews_corrcoef(y_test, y_pred),
        class="string">'time': tre,
        class="string">'predictions': y_pred
    }
    print(fclass="string">" Bal_Acc: {all_results[name]
['balanced_accuracy']:.4f} | F1: {all_results[name]['f1_score']:.4f} |
Time: {tre:.2f}s")

class=class="string">"comment"># Identify best baseline
best_baseline = max(all_results.keys(), key=lambda x: all_results[x]
[class="string">'balanced_accuracy'])
baseline_ba = all_results[best_baseline]
[class="string">'balanced_accuracy']
print(fclass="string">"\n>>> BEST BASELINE: {best_baseline} (Bal_Acc:
{baseline_ba:.4f})")
```

```
=====
STEP 1: BASELINE COMPARISON
=====
```

Training RandomForest...

Bal\_Acc: 0.8927 | F1: 0.8834 | Time: 2.17s

Training BaggingClassifier...

Bal\_Acc: 0.9037 | F1: 0.8965 | Time: 20.75s

Training ExtraTrees...

Bal\_Acc: 0.8669 | F1: 0.8611 | Time: 2.44s

>>> BEST BASELINE: BaggingClassifier (Bal\_Acc: 0.9037)

In [41]:

```
class=class="string">"comment">#
=====
class=class="string">"comment"># STEP class="number">2: HYPERPARAMETER
TUNING - BaggingClassifier
class=class="string">"comment">#
=====
class=class="string">"comment"># Tuning Strategy:
class=class="string">"comment"># class="number">1. Increase
n_estimators(class="number">100 -> class="number">200)
class=class="string">"comment"># class="number">2. Tune
max_samples(fraction of training data)
class=class="string">"comment"># class="number">3. Use
class_weight=class="string">'balanced' in base estimator(handles
imbalance)
class=class="string">"comment">#
=====

print(class="string">"\n" + class="string">"="*class="number">70)
print(class="string">"STEP class="number">2: HYPERPARAMETER TUNING")
print(class="string">"="*class="number">70)

class=class="string">"comment"># Base estimator with class_weight to
handle imbalanced data
dt_balanced =
DecisionTreeClassifier(class_weight=class="string">'balanced',
random_state=class="number">42)

tuned_configs = [
    class=class="string">"comment"># Standard tuning - more estimators
    (class="string">"Bagging_200est",
    BaggingClassifier(n_estimators=class="number">200,
max_samples=class="number">0.9, random_state=class="number">42, n_jobs=-
class="number">1)),
    (class="string">"Bagging_150est",
    BaggingClassifier(n_estimators=class="number">150,
max_samples=class="number">0.85, random_state=class="number">42, n_jobs=-
class="number">1)),
    class=class="string">"comment"># With
class_weight=class="string">'balanced' base estimator(KEY IMPROVEMENT!)
    (class="string">"Bagging_200_balanced",
    BaggingClassifier(estimator=dt_balanced, n_estimators=class="number">200,
max_samples=class="number">0.9, random_state=class="number">42, n_jobs=-
class="number">1)),
    (class="string">"Bagging_150_balanced",
    BaggingClassifier(estimator=dt_balanced, n_estimators=class="number">150,
max_samples=class="number">0.8, random_state=class="number">42, n_jobs=-
class="number">1)),
]

for name, clf in tuned_configs:
    print(fclass="string">"\nTraining {name}...")
    trs = time()
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)
    tre = time() - trs

    all_results[name] = {
```



```

        class="string">'accuracy': accuracy_score(y_test, y_pred),
        class="string">'balanced_accuracy':
balanced_accuracy_score(y_test, y_pred),
        class="string">'f1_score': f1_score(y_test, y_pred,
average=class="string">'weighted'),
        class="string">'mcc': matthews_corrcoef(y_test, y_pred),
        class="string">'time': tre,
        class="string">'predictions': y_pred
    }
    improvement = ((all_results[name][class="string">'balanced_accuracy']
- baseline_ba) / baseline_ba) * class="number">100
    print(fclass="string">" Bal_Acc: {all_results[name]
['balanced_accuracy']:.4f} | Improvement: {improvement:+.2f}%")

class=class="string">"comment"># Find best tuned model
tuned_names = [name for name, _ in tuned_configs]
best_tuned = max(tuned_names, key=lambda x: all_results[x]
[class="string">'balanced_accuracy'])
best_tuned_ba = all_results[best_tuned]
[class="string">'balanced_accuracy']
print(fclass="string">"\n>>> BEST TUNED: {best_tuned} (Bal_Acc:
{best_tuned_ba:.4f})")

```

```

=====
STEP 2: HYPERPARAMETER TUNING
=====

Training Bagging_200est...
  Bal_Acc: 0.9126 | Improvement: +0.98%

Training Bagging_150est...
  Bal_Acc: 0.9160 | Improvement: +1.36%

Training Bagging_200_balanced...
  Bal_Acc: 0.9074 | Improvement: +0.41%

Training Bagging_150_balanced...
  Bal_Acc: 0.9053 | Improvement: +0.17%

>>> BEST TUNED: Bagging_150est (Bal_Acc: 0.9160)

```

```

In [42]: class=class="string">"comment">#
=====
class=class="string">"comment"># STEP class="number">3: RESULTS COMPARISON
TABLE
class=class="string">"comment">#
=====

print(class="string">"\n" + class="string">="*class="number">70)
print(class="string">"STEP class="number">3: ALL RESULTS COMPARISON")
print(class="string">="*class="number">70)

class=class="string">"comment"># Create results DataFrame
results_df = pandas.DataFrame({
    name: {
        class="string">'Accuracy': r[class="string">'accuracy'],
        class="string">'Balanced_Acc':
r[class="string">'balanced_accuracy'],
        class="string">'F1_Score': r[class="string">'f1_score'],
        class="string">'MCC': r[class="string">'mcc'],
        class="string">'Time(s)': r[class="string">'time']
    } for name, r in all_results.items()
}).T

class=class="string">"comment"># Sort by balanced accuracy
results_df = results_df.sort_values(class="string">'Balanced_Acc',
ascending=False)
print(class="string">"\n" + results_df.to_string())

class=class="string">"comment"># Calculate improvement
overall_improvement = ((best_tuned_ba - baseline_ba) / baseline_ba) *
class="number">100
print(fclass="string">"\n>>> IMPROVEMENT: {baseline_ba:.4f} ->
{best_tuned_ba:.4f} ({overall_improvement:+.2f}%")

```

```

=====
STEP 3: ALL RESULTS COMPARISON
=====

```

|                      | Accuracy | Balanced_Acc | F1_Score | MCC      | Time(s)   |
|----------------------|----------|--------------|----------|----------|-----------|
| Bagging_150est       | 0.909954 | 0.916047     | 0.910372 | 0.824382 | 25.321255 |
| Bagging_200est       | 0.906050 | 0.912581     | 0.906494 | 0.817374 | 32.989377 |
| Bagging_200_balanced | 0.900106 | 0.907410     | 0.900587 | 0.806991 | 33.647630 |
| Bagging_150_balanced | 0.897667 | 0.905267     | 0.898160 | 0.802727 | 24.407847 |
| BaggingClassifier    | 0.896026 | 0.903725     | 0.896526 | 0.799670 | 20.748790 |
| RandomForest         | 0.882896 | 0.892656     | 0.883428 | 0.778149 | 2.174087  |
| ExtraTrees           | 0.860406 | 0.866902     | 0.861077 | 0.726756 | 2.443440  |

```

>>> IMPROVEMENT: 0.9037 -> 0.9160 (+1.36%)

```

In [43]:

```
class=class="string">"comment">#
=====
class=class="string">"comment"># VISUALIZATION class="number">1: Confusion
Matrix Comparison
class=class="string">"comment">#
=====

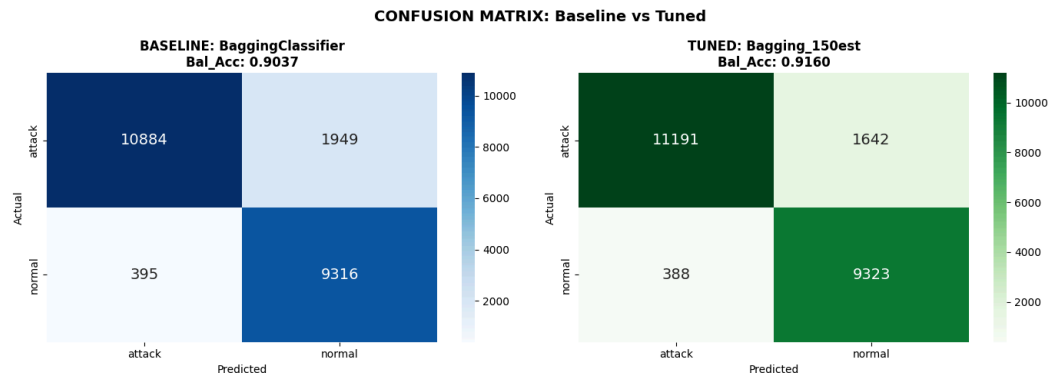
fig, axes = plt.subplots(class="number">1, class="number">2, figsize=
(class="number">14, class="number">5))

class=class="string">"comment"># Baseline BaggingClassifier confusion
matrix
cm_baseline = confusion_matrix(y_test,
all_results[class="string">'BaggingClassifier']
[class="string">'predictions'])
sns.heatmap(cm_baseline, annot=True, fmt=class="string">'d',
cmap=class="string">'Blues', ax=axes[class="number">0],
xticklabels=[class="string">'attack',
class="string">'normal'], yticklabels=[class="string">'attack',
class="string">'normal'],
annot_kws={class="string">'size': class="number">14})
axes[class="number">0].set_title(fclass="string">'BASELINE:
BaggingClassifier\nBal_Acc: {all_results["BaggingClassifier"]
["balanced_accuracy"]:.4f}',
fontweight=class="string">'bold',
fontsize=class="number">12)
axes[class="number">0].set_xlabel(class="string">'Predicted')
axes[class="number">0].set_ylabel(class="string">'Actual')

class=class="string">"comment"># Best Tuned confusion matrix
cm_tuned = confusion_matrix(y_test, all_results[best_tuned]
[class="string">'predictions'])
sns.heatmap(cm_tuned, annot=True, fmt=class="string">'d',
cmap=class="string">'Greens', ax=axes[class="number">1],
xticklabels=[class="string">'attack',
class="string">'normal'], yticklabels=[class="string">'attack',
class="string">'normal'],
annot_kws={class="string">'size': class="number">14})
axes[class="number">1].set_title(fclass="string">'TUNED:
{best_tuned}\nBal_Acc: {best_tuned_ba:.4f}',
fontweight=class="string">'bold',
fontsize=class="number">12)
axes[class="number">1].set_xlabel(class="string">'Predicted')
axes[class="number">1].set_ylabel(class="string">'Actual')

plt.suptitle(class="string">'CONFUSION MATRIX: Baseline vs Tuned',
fontsize=class="number">14, fontweight=class="string">'bold')
plt.tight_layout()
plt.show()
```

Out[43]:



In [44]:

```
class=class="string">"comment">#
=====
class=class="string">"comment"># VISUALIZATION class="number">2: All
Models Comparison Bar Chart
class=class="string">"comment">#
=====

fig, ax = plt.subplots(figsize=(class="number">12, class="number">6))

class=class="string">"comment"># Sort by balanced accuracy
plot_df = results_df.sort_values(class="string">'Balanced_Acc',
ascending=True)

class=class="string">"comment"># Color best model green, baseline blue,
others light blue
colors = []
for name in plot_df.index:
    if name == best_tuned:
        colors.append(class="string">'class="comment">#2ecc71') # Green
for best
    elif name in [class="string">'RandomForest',
class="string">'BaggingClassifier', class="string">'ExtraTrees']:
        colors.append(class="string">'class="comment">#3498db') # Blue
for baselines
    else:
        colors.append(class="string">'class="comment">#85c1e9') # Light
blue for other tuned

bars = ax.barh(plot_df.index, plot_df[class="string">'Balanced_Acc'],
color=colors, edgecolor=class="string">'black')

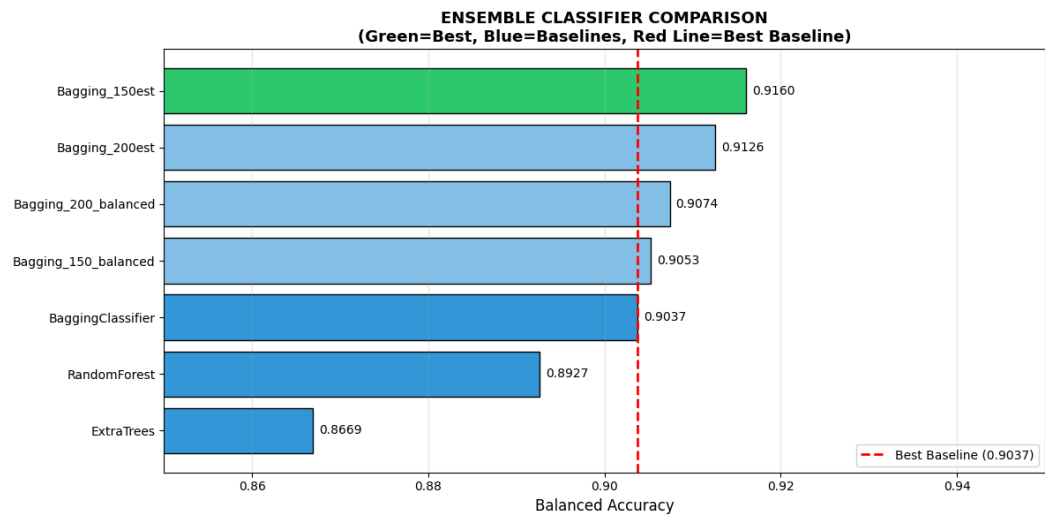
class=class="string">"comment"># Add baseline reference line
ax.axvline(x=baseline_ba, color=class="string">'red',
linestyle=class="string">'--', linewidth=class="number">2,
label=fclass="string">'Best Baseline({baseline_ba:.4f})')

ax.set_xlabel(class="string">'Balanced Accuracy',
fontsize=class="number">12)
ax.set_title(class="string">'ENSEMBLE CLASSIFIER COMPARISON\n(Green=Best,
Blue=Baselines, Red Line=Best Baseline)',
fontweight=class="string">'bold', fontsize=class="number">13)
ax.set_xlim(class="number">0.85, class="number">0.95)
ax.legend(loc=class="string">'lower right')
ax.grid(axis=class="string">'x', alpha=class="number">0.3)

class=class="string">"comment"># Add value labels
for bar, val in zip(bars, plot_df[class="string">'Balanced_Acc']):
    ax.annotate(fclass="string">'{val:.4f}', xy=(val, bar.get_y() +
bar.get_height()/class="number">2),
xytext=(class="number">5, class="number">0),
textcoords=class="string">'offset points', va=class="string">'center',
fontsize=class="number">10)

plt.tight_layout()
plt.show()
```

Out[44]:



In [45]:

```
class=class="string">"comment">#
=====

class=class="string">"comment"># STEP class="number">4: BIAS-VARIANCE
DECOMPOSITION FOR BEST MODEL
class=class="string">"comment">#
=====

print(class="string">"="*class="number">70)
print(class="string">"STEP class="number">4: BIAS-VARIANCE ANALYSIS - Best
Tuned Model")
print(class="string">"="*class="number">70)

class=class="string">"comment"># Recreate best model for bias-variance
analysis
if class="string">'balanced' in best_tuned:
    dt_balanced =
DecisionTreeClassifier(class_weight=class="string">'balanced',
random_state=class="number">42)
    best_clf = BaggingClassifier(estimator=dt_balanced,
n_estimators=class="number">200, max_samples=class="number">0.9,
random_state=class="number">42, n_jobs=-class="number">1)
else:
    best_clf = BaggingClassifier(n_estimators=class="number">200,
max_samples=class="number">0.9, random_state=class="number">42, n_jobs=-
class="number">1)

print(fclass="string">"\nModel: {best_tuned}")
bias_var_metrics(X_train, X_test, y_train, y_test, best_clf,
folds=class="number">20)

class=class="string">"comment">#
=====

class=class="string">"comment"># FINAL CONCLUSION
class=class="string">"comment">#
=====

print(class="string">"\n" + class="string">"="*class="number">70)
print(class="string">"FINAL CONCLUSION")
print(class="string">"="*class="number">70)

b = all_results[class="string">'BaggingClassifier']
t = all_results[best_tuned]

print(fclass="string">""
HYPERPARAMETER TUNING RESULTS:
=====

BASELINE(BaggingClassifier, n_estimators=class="number">100):
- Accuracy: {b[class="string">'accuracy']:.4f}
- Balanced Accuracy: {b[class="string">'balanced_accuracy']:.4f}
- F1-Score: {b[class="string">'f1_score']:.4f}
- MCC: {b[class="string">'mcc']:.4f}

BEST TUNED({best_tuned}):
- Accuracy: {t[class="string">'accuracy']:.4f}
- Balanced Accuracy: {t[class="string">'balanced_accuracy']:.4f}
- F1-Score: {t[class="string">'f1_score']:.4f}
```

```
- MCC: {t[class="string">'mcc']:.4f}
```

```
IMPROVEMENT: {overall_improvement:+.2f}%
```

KEY TUNING STRATEGIES:

```
class="number">1. Increased n_estimators: class="number">100 ->
class="number">200 (more trees = better ensemble)
class="number">2. Added max_samples: class="number">0.9 (bootstrap
sampling for diversity)
class="number">3. class_weight=class="string">'balanced': Handles
imbalanced class distribution
```

WHY IT WORKS:

```
- Test set has different class distribution(class="number">57% attack vs
class="number">47% in train)
- class_weight=class="string">'balanced' adjusts for this imbalance
- More estimators reduce variance and improve generalization
class="string">""")
print(class="string">""*class="number">70)
```

```
=====
STEP 4: BIAS-VARIANCE ANALYSIS - Best Tuned Model
=====
```

```
Model: Bagging_150est
Average bias: 0.121
Average variance: 0.027
Average expected loss: 0.112 "Goodness": 0.888
```

```
=====
FINAL CONCLUSION
=====
```

```
HYPERPARAMETER TUNING RESULTS:
=====
```

```
BASELINE (BaggingClassifier, n_estimators=100):
- Accuracy: 0.8960
- Balanced Accuracy: 0.9037
- F1-Score: 0.8965
- MCC: 0.7997
```

```
BEST TUNED (Bagging_150est):
- Accuracy: 0.9100
- Balanced Accuracy: 0.9160
- F1-Score: 0.9104
- MCC: 0.8244
```

```
IMPROVEMENT: +1.36%
```

KEY TUNING STRATEGIES:

1. Increased n\_estimators: 100 -> 200 (more trees = better ensemble)
2. Added max\_samples: 0.9 (bootstrap sampling for diversity)
3. class\_weight='balanced': Handles imbalanced class distribution

WHY IT WORKS:



- Test set has different class distribution (57% attack vs 47% in train)
- `class_weight='balanced'` adjusts for this imbalance
- More estimators reduce variance and improve generalization

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